

RUBBER ACCELERATORS

_____ Name
_____ Date

What are these?

Rubber accelerators and antioxidants are chemicals used in the manufacture of both natural latex and synthetic rubbers. Many of these agents can cause allergic dermatitis. This type allergy is different from Latex Allergy, in which potentially dangerous *immediate* swelling occurs on contact with natural rubber made from latex tree sap.

Where might they be found?

rubber gloves, nitrile gloves
rubber sandals, slippers, soles
rubber shoes, insoles, boots
rubber hip waders, gators
elastic in clothing, socks, girdle
support stockings
non-slip waistband, elastics
brassiere cups
rubber undergarments, panties
slimming suits, sauna suits
swimwear, swim cap, goggles
wetsuit, diving gear
condoms, diaphragms
rubber vibrator, rubber dropper
rubber pillow, mattress, sheet
cosmetic applicator, lash curler
ear plugs, headphones
hairbrush handle, hairnet
toothbrush rubber handle
dentures, dental dams
baby bottle nipples, pacifiers
balloons, rubber balls, toys
adhesive tape, bunion pad
hot water bottle, urinal
medical devices, stethoscopes
intravenous port, stoppers
tourniquet, syringe plunger
blood pressure cuff
surgical mask, CPAP strapping
dialysis equipment, catheter
orthopedic braces, supports
rubber prosthesis, splint, mask
rubber bands, i-Pod band
pen handle, pencil grip, eraser
mouse pad, wrist support
glue, envelopes, stamps
rubber bank note counter

kitchen utensil rubber handles
rubber mats, carpet backing or underlay pad
racquet handle, club handle
hockey pads, soccer shin pad, foam shoulder pad
rubber in upholstered furniture
rubber cement for shoe uppers and linings
rubber hose, rubber tubing, seals, cables
steering wheel, bicycle handle
auto hoses, gaskets, fan belts
shock absorbers, springs, tires, inner tubes
tool handles, utensil handles
duct tape, electrical cords
weather stripping
safety aprons, goggles, respirators, gas masks
neoprene and isoprene rubber
conveyor belts
fuel tank lining
synthetic resins in textile/dye industry
microscope eyepiece

How to avoid them:

Most rubber products contain rubber accelerators. Avoid direct contact with all rubber. Use protective fabric, leather, or plastic to handle dry rubber. Wear heavy duty vinyl gloves to protect against workplace contact with these chemicals.

You were patch tested with these accelerators:

- ♦ **Thiurams** (tetramethylthiuram monosulfide, tetramethylthiuram disulfide, tetraethylthiuram disulfide, dipentamethylene thiuram disulfide),
- ♦ **Carbamates** (zinc diethyldithiocarbamate, diphenylguanidine, zinc dibutyldithiocarbamate),
- ♦ **Mixed Dialkyl Thioureas** (diethylthiourea and dibutylthiourea), and
- ♦ **Benzothiazoles** (morpholinyl mercaptobenzo-thiazole, mercaptobenzothiazole, N-cyclohexyl-2-benzothiazyl sulfenamide, dibenzothiazyl disulfide). *(continued)*

Non-rubber uses:

These chemicals have other uses that may be important in your exposure. Check the Material Safety Data Sheet (MSDS) at work for them.

Thiuram chemicals are used in germicides, pesticides, seed disinfectant, fungicides, insect repellants, animal repellants, wood and paint preservatives, plastics, putty, and as a bacteriostat in soaps, greases, shampoos, and oils. The medicine disulfuram (*Antabuse*) is a thiuram.

Carba chemicals are found in animal repellants, pesticides, herbicides, fungicides, slime retardants, disinfectants, and some shampoos, soaps, cements, sealants, and adhesives.

Thioureas are present in diazo copy paper, photography fixative, polyvinyl chloride adhesive tapes, commercial paint and glue remover, acidic detergent, silver polish dip, disinfectants, pesticides, fungicides, phonecard thermocoating, metal anti-corrosion agent, stain remover, and manufacture of resin and Neoprene.

Benzothiazoles have uses in greases, cutting fluids, antifreeze, anticorrosives, cements, adhesives, detergents, veterinary products such as tick and flea powders and sprays, fungicides, slimicides, tree spray, photographic film emulsions, pottery molds and releasing fluid, and paints.

Replace Rubber Items with these Alternatives:

Spandex	polyvinyl chloride	wood
silicone	buttons (for elastic)	acrylic
plastic	drawstrings (for elastic)	vinyl
fabric	polyethylene foam	Mylar

NOTE: Any socks or stockings worn with shoes that caused the allergic reaction may have gotten contaminated. Washing may not remove the allergen.

These are FREE of Rubber Accelerators:

Household gloves

- ♦ Best N-DEX NightHawk Defender 800-241-0323
- ♦ Allerderm Heavy Duty Vinyl 800-365-6868

Exam gloves (all Nitrile) *

- ♦ Ansell Microflex Sensation
- ♦ Ansell Micro-Touch NitraFree
- ♦ Cardinal Low Dermatitis Potential
- ♦ Dynarex Tillotson True Advantage
- ♦ Hourglass International HandPRO FreeStyle1100
- ♦ Sempermed SemperSure
- ♦ SmartPractice Reflection Sapphire Sensitive PF Violet Blue
- ♦ SmartPractice Ultimate N-DEX Free Powder Free

Sterile surgical gloves *

- ♦ Ansell GAMMEX Non-Latex Neoprene
- ♦ Ansell GAMMEX Non-Latex Sensitive Synthetic Sensoprene
- ♦ Medline DermAssure Green Powder-Free Neoprene
- ♦ Molnlycke Health Care Biogel NeoDerm Neoprene

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Chemical protective gloves

- ♦ 4H/Silvershield
- ♦ Barrier Chemical Protective

Condoms

- ♦ Trojan Naturalamb
- ♦ Durex Avanti polyurethane

Diapers

- ♦ Luvs, Pampers

Underwear—with no elastic waist, leg bands, straps

- ♦ Jockey, Dorlastan, Spandex (many brands)

Socks—with no elastic component **Shoes**—completely leather with non-rubber sole and insole (e.g. cork, felt)

Sports equipment free of Neoprene—Cropper Medical, Ashland, OR; Bauerfiend USA, Wichita, KS; Speedo, Los Angeles, CA; TYR Sport, Huntington, CA; Skyline Northwest, Lake Oswego, OR; Shock Doctor, Plymouth, MN; McKeon Products, Warren, MI; E.A.R. Inc., Boulder, CO; All-Star Sports Equipment, Shirley, MA